SYSTEM ANALYSIS DOCUMENTATION (SAD)

UMHackathon 2026

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1. Introduction:

Content Buddy *is an AI-native content strategy engine designed to solve the rigidity and technical barriers of professional content creation. This document details the technical design and architectural decisions that enable style cloning, automated compliance, and iterative video planning.*

1.1. Purpose:

The system analysis document highlights the technical scope and design-related decisions behind the Content Buddy development.

So, the key elements of the system that have been covered are as follows:

1. ***Architecture****:*
    a. *It's built as a stateful agentic workflow system that centralises the fragmented steps of content creation into a unified environment.*
    b. *It's client-server model utilizing a Java Spring Boot backend and a React 19 frontend.*
2. ***Data Flows:***
    a. *The journey of user-provided URLs and prompts through the Java middleware to the Z.ai, persisting in a shared JSON state.*
    b. *The transition of raw URLs into extracted style DNA and eventually into structured strategy JSON.*
3. ***Model Process:***
    a. *The model shows the end-to-end workflow of the core features: idea and style suggestion, script and storyboard generation, and compliance auditing.*
4. ***Role of Reference:***
    a. *A guide for developers and testers to maintain system modularity and API contracts.*

1.2. Background:

Social media plays a powerful role in influencing behavior and has become essential for modern marketing, yet the content creation process remains highly fragmented and inefficient. Creators must juggle multiple platforms for trends, music, and scripting, leading to wasted time, reduced productivity, inconsistent branding, and creative burnout. At the same time, they face a "compliance minefield," where misunderstanding platform rules can result in low reach, content removal, or account penalties. Although many aim to turn ideas into viral content, a growing creation bottleneck persists, further worsened by existing AI tools that lack brand memory and produce generic, inconsistent outputs—ultimately causing a strategic engagement gap where content fails to authentically connect with audiences.

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1. *Previous Version:*
    a. *Creators had to manually browse social media or use paid SEO tools to find a topic.*
    b. *The generic chatbot AI didn't know the user's previous videos, so the script and description often felt robotic and off-brand.*
    c. *Standard AI tools only provide text, so the creator still needs to manually create the storyboard.*
    d. *Before posting, the creator had to manually read through social media guidelines to ensure they were not shadowbanned.*
2. *Changes in Major Architectural Components*
    a. *Instead of requiring users to manage multiple independent systems, Content Buddy has combined these into a single, simple system.*
    b. *Content Buddy introduces an agentic co-pilot layer that allows stateful refinement of content.*
3. *New capabilities that are only introduced in this*
    a. *Content Buddy introduces Agentic Reasoning, dual-perspective scripts (Actor/Director), and automated platform-specific risk mitigation.*
    b. *Real-time visual storyboarding via AI sketches and automated shadowban risk detection.*

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1.3. Target Stakeholder

Stakeholders	***Roles***	***Expectations***
Business & Product Teams *(Founders, PMs, Entrepreneurs)*	*Utilise the platform to create marketing strategies and "Product Demo" scripts for early-stage validation.*	*Rapid transition from product concept to promotional content without high agency costs.*
Solo Professionals *(Real Estate Agents, Fitness Coaches)*	*Act as the "on-camera" talent; use the Actor/Director scripts to film professional-grade industry content.*	*Elimination of "planning paralysis" through clear visual and verbal instructions for filming.*
E-commerce & SMEs *(Sellers, Small Business Owners)*	*Generate trend-aligned ad content and captions to drive sales for products/services.*	*High-conversion captions and Guardian checks to ensure ads aren't shadowbanned on TikTok/IG.*
Content Creators *(Influencers, UI/UX Designers)*	*Input brand URLs to "clone" their unique style and automate high-volume content scheduling*	*Consistent brand voice across all outputs and automated "Social SEO" optimisation.*
Student Organizations & Campus Clubs	*Manage social media presence for university events and community engagement.*	*A unified, easy-to-use interface that doesn't require professional multimedia training.*
Development Team	*Build, maintain, and scale the 3-Tier Agentic System and the Heartbeat (`data.json`).*	*Defined scope, the request and response formats, and coverage of edge cases*
QA Team	*Validate the system behaviour, ensuring platform rules and "Voice Cloning" are accurate.*	*Defined scope, clear request/response formats for the GLM-5.1 API, and edge-case coverage.*

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2. System Architecture & Design

2.1. High-Level Architecture

2.1.1. Overview:

Type	Details
System	*Web-Based Agentic Application*
Architecture	*Stateful Agentic Workflow System*

Content Buddy is structured as a client-server system where the React Frontend handles user interactions and state visualization. The Spring Boot Backend acts as an orchestration layer, communicating with the Z AI API for reasoning and Pollinations AI for storyboard sketches.

2.1.2 LLM as Service Layer

In the Content Buddy architecture, the Z AI GLM-5.1 flash is integrated as the Core Reasoning Service Layer. It acts as the "Central Intelligence" that processes data provided by the Java Orchestrator to generate structured, actionable multimedia strategies.

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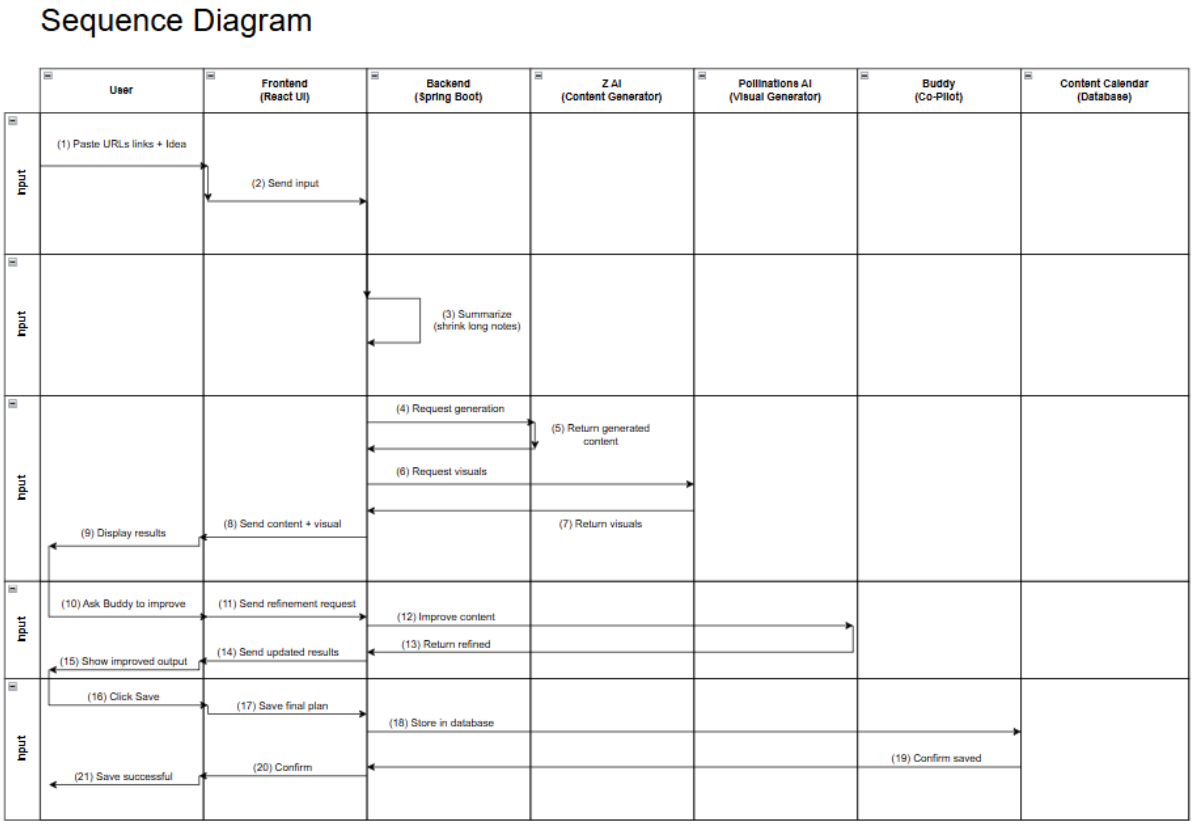
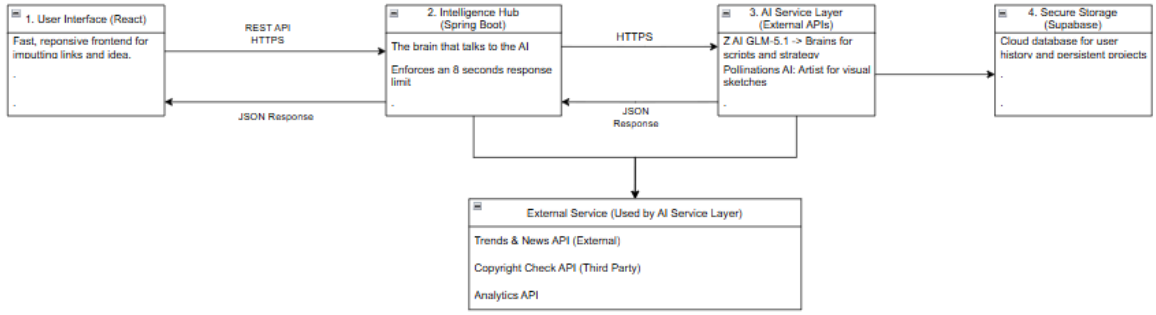
2.1.3 Dependency Diagram



2.2.2. Sequence Diagram



2.2. Technological Stack

2.2.1. Frontend: React 19, Vite, Tailwind CSS v4, Framer Motion

2.2.2. Backend: Java 17, Spring Boot 3.4+, RestTemplate

2.2.3. AI Engine: Z.AI GLM

2.2.4. Database: Supabase

2.2.5. Visuals: Pollinations AI

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2.3. Key Data Flows

2.3.1. Here, we will show how data is moving through the system and how its structure is stored.

2.3.1.1. Data Flow Diagram (DFD)

a. INPUT: User URLs and Subject String
b. PROCESS 1: Style Extraction -> Backend LLM call
c. PROCESS 2: Compliance Scan -> React-based policy logic
d. OUTPUT: Structured JSON project state
e. PROCESS 3: State Persistence -> React session storage and global state for schedule

2.3.1.2. Normalized Database Schema

f. Project Data:
  i. { hook: string, sonicDna: string, script: string, storyboard: Array<Scene> }
g. User Information (Core identity):
  i. {user id: username: email: password hash: created time: last login}
h. Scene:
  i. { project id: int, scene: string, character: string, shooting: string, editing: string, dialogue: string, duration: string }
i. History table:
  i. {project id: user id: title: content json: platform: url reference: created time}

3. Functional Requirements & Scope

3.1. Minimum Viable Product:

#	*Features*	*Description*
1	*Signature Style Learning*	Examines three external videos in order to understand the overall vibe and caption types used to ensure that new ideas feel like true originals.
2	Visual Storyboarding	Creates scene visuals to prepare for filming the content and help define the overall concept.
3	Multi-Perspective Scripting	Produces both an actor version (dialogue and delivery) and a director version (camera angles, lighting, and transitions) for complete production guidance.

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4	Intelligent Content Calendar	A constant calendar icon that shows in the user interface allows users to manage the global content pipeline and visualize all upcoming titles and dates.
5	Precision Posting Notifications	Sends timely alerts to maximize reach and posting efficiency.
6	Multimodal Asset Generator	Automates the generation of an entire production package, including suggested title, popular songs for use, and optimal captions and scripts.
7	Centralized Navigation & History (Modern UI/UX)	Clean interface with persistent side, top, and bottom panels, dark/light mode, and easy access to past content and settings.
8	Compliance&Safety Engine (Copyright Reminder)	Integrated safeguard within Content Buddy, designed to analyze the selected platform and its specific policies to ensure that users remain compliant with platform policies and reduces the risk of copyright infringement or content removal.

3.2. Non-Functional Requirements (NFRs)

Quality (e.g.)	***Requirements (e.g.)***	***Implementation (e.g)***
Reliability	*State must persist if the browser refreshes.*	*Continuous writing to cloud persistent storage via Supabase on the server disk.*
Maintainability	*Logic for TikTok vs. YouTube must be separate.*	*Strategy Pattern in Java backend for platform rules.*
Token Latency	*GLM responses must appear in the UI within 8 seconds.*	*Use of GLM-5.1-flash model for high-speed inference.*
Cost Efficiency	*Minimize unnecessary API calls*	*Character truncation (500 chars) and system prompt optimization.*

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3.3. Out of Scope / Future Enhancements

a. *End-to-End Video Production: Integration with AI video tools (e.g., Sora, HeyGen) to automatically generate complete video outputs from scripts.*

b. *One-Click Publishing: Introduce a "Post Now" feature to directly upload content to social media platforms via API integration.*

c. *Real-Time Collaborative Workspace: Enable multi-user collaboration with live editing, shared links, and synchronized project environments.*

d. *In-App Creative Editor: Develop a built-in editor for users to modify scripts, storyboards, and content assets within the platform.*

e. *Global Linguistic Adaptability: Support multilingual content generation to allow users to create and strategize in multiple languages.*

f. *Premium Membership & Payment System: Implement subscription plans and secure payment gateways (e.g., Stripe) for advanced features.*

g. *High-Speed Memory (Caching System): Use technologies like Redis caching to improve response speed and reduce system costs.*

4. Monitor, Evaluation, Assumptions & Dependencies

4.1. Technical Evaluation:

4.1.1. Grayscale Rollout & A/B Testing:

4.1.1.1. Future roll out will test two different GLM system prompts to see which generates higher engagement scores.

4.1.2. Emergency Rollback (ER) & Golden Release:

4.1.2.1. In case of an API timeout or 500 error from z.ai, the system rolls back to a "Template based" script generator to ensure the user still receives an output.

4.1.3. Priority Matrix:

4.1.3.1. P1 (Critical): If cloud persistent storage via Supabase becomes corrupted, the system triggers an automatic backup restore.

4.1.3.2. P2 (High): If Z.ai latency exceeds 10s, the UI displays a "High Traffic" notification.

4.1.3.3. P3 (Medium): If schedule database failure, the system will keep data in local session state until the problem is resolved.

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4.2. Monitoring:

4.2.1. *Show the planning for your monitoring plan of the system and how it will trigger if any damage is detected of the health of the system.*

4.3. Assumptions:

a. *Users must have an active internet connection for API calls and real-time content generation.*

b. *Users must input prompts, reference links and selecting the target platform.*

c. *URLs provided by users are public and accessible.*

4.4. External Dependencies:

Tools	Purpose	Risks
Z Ai GLM API	Core reasoning engine and JSON generation.	High: Potential rate limits
Pollinations AI	Generating visual storyboard sketches.	Medium: Accuracy of visual interpretation
GitHub Actions	CI/CD pipeline automation.	Low
Supabase	User authorized and database.	Low: Can fall back to local JSON.

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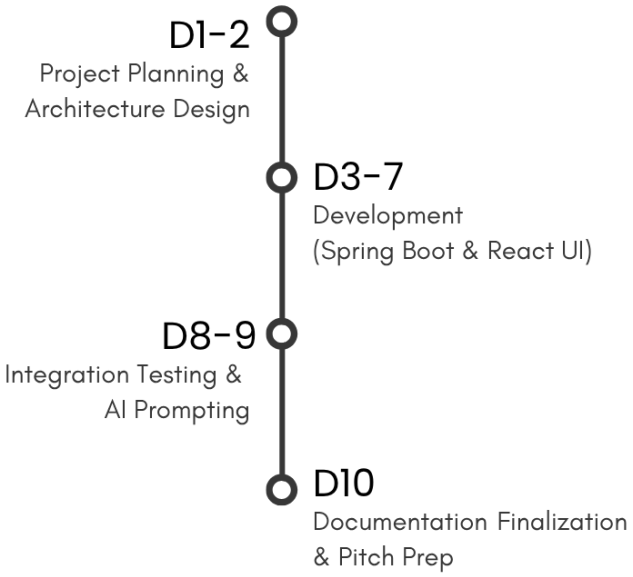
5. Project Management & Team Contributions

The duration of the Preliminary round is approximately 10 days considering the two-week sprint.

5.1. Project Timeline:



5.2. Team Members Role:

Name	*Roles*
Victoria Kew Kim Tian	*Team lead, Lead Full-stack developer*
Yow Jia Yen	*Full-stack developer, UI/UX*
Goh Ching Yee	*Database and documentation*
Ong Jing Zhi	*Documentation and slides*
Ivy Ng Xin Rou	*Documentation and video*

5.3 Recommendations:

5.2.1. *Implement Redis Caching for niche specific suggestions to reduce LLM costs*

5.2.2. *Use Websockets for the compliance scan to have real time feedback for policy compliance*

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